



UNIT 1: INTRODUCTION TO COMPUTERS

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Content:	1. Alexis Leon and Mathews Leon, "Fundamentals of Information Technology", Vikas Publication, Second edition, 2009.	No of hours
Unit - 1	Introduction to Computers: Importance of computers, characteristics of computers, classification of computers, uses of computers. Anatomy of Digital Computer: parts of computer, CPU: Control Unit and ALU. secondary storage devices, keyboards, mouse, scanners, readers, digital cameras, monitors, and printers. Operating Systems: Introduction, functions of an operating system, classification of operating systems. Introduction to Computer Problem Solving: Introduction, problem-solving aspect, top-down design.	10

- A **computer** is a machine that **performs calculations and processes information** with astonishing **speed and precision**.
- It can **handle vast amount of information** and **handle complicated problems**.

Two **principal characteristics** of a computer are:

1. It **responds to a specific set of instructions** in a well defined manner
2. It can **execute a prerecorded list of instructions**(a program)

IMPORTANCE OF COMPUTERS

- **Simplify difficult or time consuming tasks.**
- **Provide** businesses, governments , individuals and institutions with an **efficient way to manage large amount of information**
- It **performs certain tasks quickly and accurately. These tasks include:**
 1. **Solving numerical ability:** computers can solve numeric problems very quickly.
 2. **Storing and retrieving information:** People can use computers to store large quantities of information. Information stored in a computer is sometimes called as a **database**.
 3. **Creating and displaying documents and pictures:** **Word processing program** allows people to type words into a computer to **write articles, books, letters, reports** and other kinds of documents. **Computer graphics** is use of computers to **make pictures**.

CHARACTERISTICS OF COMPUTERS

Computers have the following characteristics:

1. Word length:

- **A digital computer operates on binary digits 0 and 1.** Binary digit is called a bit. Group of 8 bits is a byte.
- **Word length:** Number of bits a computer can process at a time.
- **Commonly used word lengths are 8,16,32,64.**
- **Greater the word length more powerful the computer.**

2. Speed

- Computers can calculate at very high speeds.
- **Microcomputer** can execute millions of instructions per second without any mistake.
- **Supercomputers** can operate at speeds measured in nanoseconds or in picoseconds- one thousand to one million times faster than microcomputers.

3. Storage

- Computers have **main memory and auxiliary memory systems.**
- **Auxiliary storage devices are capable of storing huge amounts of data.**
- Information that the user wants can be retrieved in few seconds.

CHARACTERISTICS OF COMPUTERS

4. Accuracy

- **Accuracy** of a computer system is **very high**.
- Errors in **hardware can occur but error detecting and correcting techniques will prevent false results**.
- In most cases errors are due to human factor rather than technological flaws.

5. Versatility

- Computers can perform activities ranging from simple calculations to navigating missiles and satellites.
- They are capable of **performing almost any task**.
- They can **communicate with other computers**.

6. Automation

- Once a task is initiated **computers can proceed on its own till its completion**.
- They execute programs in correct sequence, switch on or switch off machines at correct time etc.

7. Diligence

- **Computers perform tasks irrespective of whether it is interesting, creative, boring etc.**
- Do not have any human weaknesses

TYPES OF COMPUTERS

Microcomputer:

- Small, single user computers based on a microprocessor. They have keyboard monitors, storage devices.
- They are further classified as personal computers, workstations, portable computers etc.

Minicomputer:

- Is a multiuser computer capable of supporting ten to hundreds of users simultaneously.

Mainframe:

- Powerful multiuser computers capable of supporting hundreds of users simultaneously.
- Used by banks, airline systems etc.

Supercomputer.

- Extremely fast computer.
- Used by government agencies
- These machines are for applications requiring very large programs and huge amounts of data that must be processed quickly
- The main **difference** between a **supercomputer** and a **mainframe** is that a **supercomputer channels all its power into executing a few programs as fast as possible, whereas a mainframe uses its power to execute many programs concurrently.**

USES OF COMPUTERS

Education:

- Online learning and virtual classrooms
- Educational software and interactive learning tools
- Research and information access

Healthcare:

- Patient record management and electronic health records (EHRs)
- Diagnostic tools and medical imaging

Business:

- Data analysis and business intelligence
- Accounting and financial management

Communication:

- Email and instant messaging
- Video conferencing
- Social media and networking platforms

USES OF COMPUTERS

Entertainment:

- Gaming
- Streaming movies, music, and videos

Government:

- E-governance and online public services
- Data management
- Defense and security systems

Finance:

- Online banking and financial transactions
- Stock trading and market analysis
- Risk management and fraud detection

Personal Use:

- Personal finance management
- Online shopping and digital payments

THANK YOU😊

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